

NYC 311 & Weather Correlation Dashboard

Data Analysis Report

Prepared by: Alison Hartshorn

Data Sources: NYC Open Data (311 Service Requests) · NOAA Global Historical Climatology Network

Pipeline: Python · PostgreSQL · Power BI · Windows Task Scheduler

Research Question

How do weather conditions, temperature, precipitation, and wind, drive the volume and type of 311 complaints filed across New York City boroughs, and can patterns in the data inform more proactive city resource allocation?

Executive Summary

This report analyzes 941,876 NYC 311 service complaints filed between February and April 2026 and correlates them with daily weather observations from four NOAA stations across New York City. The data was ingested via REST APIs, cleaned through a multi-stage PostgreSQL ETL pipeline, and visualized in an interactive Power BI dashboard with automated daily refresh.

Five key findings emerged from the analysis:

1. Illegal parking is the single most common complaint across all five boroughs, accounting for over 140,000 reports in 90 days.
2. Cool weather (5–18°C) generates more complaints than any other temperature range — driven primarily by heating failures and snow/ice conditions.
3. Brooklyn consistently leads all boroughs in complaint volume, filing 30% more complaints than Queens despite a similar population.
4. Food establishment and cannabis retailer complaints take the longest to resolve, averaging over 1,000 hours (42+ days) to close.
5. Tuesday is the busiest complaint day citywide, while weekends show a consistent 10–15% drop in volume.

These findings suggest opportunities for more proactive resource allocation; particularly around parking enforcement, winter heating preparedness, and complaint resolution SLA enforcement for food-related agencies.

Data & Methodology

Data Sources

SOURCE	DATASET	COVERAGE
NYC OPEN DATA	311 Service Requests (erm2-nwe9)	Feb – Apr 2026 · ~5K new rows/day
NOAA CDO API	GHCND Daily Summaries — TMAX, TMIN, PRCP, SNOW, AWND	4 stations · Central Park, JFK, LaGuardia, Newark

Data Quality & Cleaning

The raw 311 dataset is intentionally messy. It is real civic data with no quality standards enforced at collection. The ETL pipeline addressed the following issues:

1. Borough normalization: values like 'MANHATTAN', 'MN', and 'manhattan' standardized to a consistent format using SQL CASE logic

2. Deduplication: 40,000+ duplicate submissions removed using ROW_NUMBER() window functions, keeping the most recent ingestion of each unique_key
3. Coordinate validation: latitude/longitude values validated against NYC's geographic bounding box. Invalid values set to NULL.
4. Zip code validation: regex filter requiring exactly 5 digits. The malformed values were discarded.
5. Temporal paradoxes: 0 records where closed_date preceded created_date were flagged and excluded from resolution time calculations
6. Weather borough approximation: no GHCND stations exist for Bronx or Brooklyn. It was approximated using Manhattan (Bronx) and Queens (Brooklyn) data

QUALITY METRIC	TARGET	ACHIEVED
NULL/UNKNOWN BOROUGH RATE	< 1%	0.1% ✓
WEATHER JOIN RATE (VALID BOROUGH)	> 95%	96.1% ✓
CLEAN ROWS LOADED TO MART	—	941,876 ✓
TEMPORAL PARADOXES (CLOSED BEFORE OPENED)	0	0 ✓

Key Findings

Finding 1 — Complaint Volume by Borough

Brooklyn generates the most complaints of any borough, followed by Queens and the Bronx. Staten Island, while the least populous borough, files roughly one-sixth of Brooklyn's complaint volume.

	BOROUGH	COMPLAINTS	SHARE OF TOTAL
1	Brooklyn	288,237	30.6%
2	Queens	233,611	24.8%
3	Bronx	192,462	20.4%
4	Manhattan	180,059	19.1%
5	Staten Island	46,543	4.9%

Brooklyn's outsized complaint volume, 30% more than Queens despite a comparable population, may reflect a combination of higher residential density, older housing stock prone to heating and maintenance failures, and greater awareness of the 311 system among residents.

Finding 2 — Top Complaint Types

Illegal parking dominates complaint volume citywide, accounting for more complaints than the next two types combined. Heat and hot water complaints rank second, reflecting the aging heating infrastructure in NYC residential buildings.

	COMPLAINT TYPE	TOTAL COMPLAINTS
1	Illegal Parking	148,000+
2	Heat/Hot Water	110,000+
3	Noise — Residential	87,000+

4	Blocked Driveway	48,914
5	Street Condition	~47,000
6	Snow or Ice	~30,000
7	Unsanitary Condition	~28,000
8	Noise — Street/Sidewalk	~26,000
9	Plumbing	~24,000
10	Traffic Signal Condition	~22,000

Finding 3 — Weather & Complaint Volume

Contrary to the intuitive assumption that hot weather drives the most complaints, cool weather (5–18°C) generated the highest complaint volume during the analysis period. This is primarily driven by two factors: the frequency of cool days in a February–April window, and the concentration of heating failure complaints during colder months.

TEMPERATURE RANGE	COMPLAINT VOLUME
COOL (5–18°C)	Highest — heating complaints, winter conditions
COLD (BELOW 5°C)	High — snow/ice, heating failures peak
WARM (18–27°C)	Moderate — noise, parking dominate
HOT (27–32°C)	Lower — heat complaints emerge but less frequent
EXTREME HEAT (32°C+)	Lowest — rare in this time window

Dry days see significantly higher total complaint volume than rainy days. This is largely a frequency effect (more dry days exist) rather than a causal relationship. Per-day complaint rates during heavy rain events are worth further investigation.

Finding 4 — Resolution Times

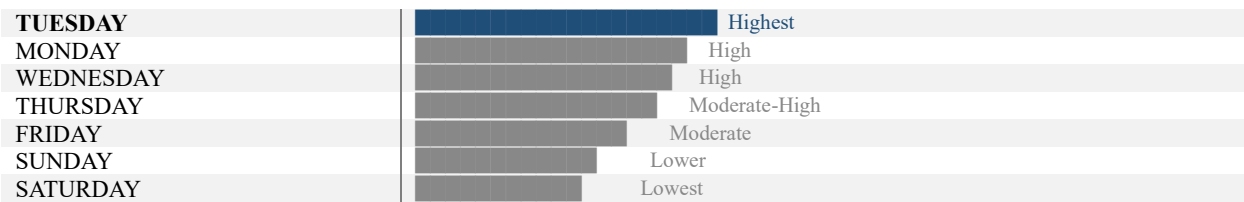
Resolution time varies dramatically by complaint type. The slowest categories involve regulatory inspections and licensing. These types that require agency coordination beyond a single department.

COMPLAINT TYPE	AVG HOURS	AVG DAYS
FOOD ESTABLISHMENT	~1,050 hrs	~44 days
CANNABIS RETAILER	~1,040 hrs	~43 days
SMOKING OR VAPING	~820 hrs	~34 days
BOILERS	~740 hrs	~31 days
BUILDING/USE	~430 hrs	~18 days
APPLIANCE	~380 hrs	~16 days
ILLEGAL PARKING	< 48 hrs	< 2 days

Finding 5 — Day of Week Patterns

Complaint volume follows a consistent weekly pattern with Tuesday as the peak day and weekends showing lower volume. This pattern holds across all five boroughs and most complaint types, suggesting institutional rhythms (business operations, city agency staffing) influence filing behavior.

DAY	RELATIVE VOLUME
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Recommendations

Operational Recommendations

1. Parking Enforcement Staffing
 - a. Increase Tuesday Coverage: Illegal parking peaks on Tuesdays. Shifting a portion of parking enforcement resources to Tuesday morning shifts could reduce complaint black log and improve resolution times for the #1 complaint type citywide.
2. Winter Heating Preparedness
 - a. Pre-position inspectors in Oct-Nov: Heat/Hot Water is the #2 complaint and spikes sharply during cool weather. Proactive boiler inspections in Brooklyn and the Bronx before winter onset could reduce the reactive complaint load during peak cold months.
3. Food Establishment SLA Reform
 - a. Target: reduce from 44 days to 21 days: Food establishment complaints average 44 days to resolve which is the slowest of any category. A dedicated fast-track inspection unit or inter-agency coordination protocol could cut resolution time significantly.
4. Brooklyn Resource Allocation
 - a. 30% more complaints than Queens: Brooklyn consistently generates the highest complaint volume despite similar population to Queens. Additional 311 call center capacity and field inspector resources targeted at Brooklyn could meaningfully reduce citywide backlog.

Analytical Next Steps

1. Extend the analysis window to 12 months to capture summer heat complaint patterns (Heat/Hot Water, air conditioning, outdoor noise) which are absent from this Feb–Apr dataset
2. Add US Census demographic data by zip code to explore whether complaint rates and resolution times vary systematically across income levels and racial demographics
3. Build a Prophet time-series forecasting model to predict complaint volume by borough 7 days ahead using NOAA weather forecasts enabling proactive rather than reactive resource deployment
4. Analyze per-capita complaint rates rather than absolute counts to fairly compare boroughs of different population sizes
5. Investigate the cannabis retailer resolution time anomaly, 43-day average, suggests possible regulatory bottleneck worth exploring with NYPD and city licensing data

Pipeline & Dashboard Notes

Pipeline Architecture

The data pipeline runs automatically every morning at 5:00 AM via Windows Task Scheduler, with the Power BI dashboard refreshing at 6:00 AM. All pipeline runs are logged to a pipeline_log table, which is surfaced in the Pipeline Health tab of the dashboard.

TOTAL RAW ROWS INGESTED	981,810
ROWS AFTER CLEANING & DEDUPLICATION	941,876
ROWS REMOVED (DUPLICATES / BAD DATES)	~39,934
WEATHER OBSERVATIONS LOADED	3,519
MART TABLE ROWS (AGGREGATED)	38,515
WEATHER JOIN RATE (VALID BOROUGHES)	96.1%
PIPELINE RUNTIME (INITIAL 90-DAY BACKFILL)	~19 minutes
PIPELINE RUNTIME (DAILY INCREMENTAL)	~2 minutes

Dashboard Access

The [interactive Power BI dashboard](#) is publicly available — no account required: